Day 14 (20/11/2025)

1) What are functions?

function is a block of reusable code that performs a Specific task.

It helps in breaking a large program in to small managable parts.

2) why?

* increase code reusability.
* improves readability & modularity.
* easier debugging & testing.

Types of function:

1) built in function -> provided by python
print(), input(), lower(), len().. etc.
2) user defined functions -> Created by the programmer using def
3) lambda functions -> anonymous, single line functions
Created with lambda.

User defined functions-

```
def fun_name(parameters name):
    statements
    return value
```

Calling the function

```
fun_name()
```

Different methods of user defined functions-

1) function without input & without return
2) function with input & without return
3) function without input & with return
4) function with input & with return

1) function without input & without return.

Syntax -

```
def fun_name():
        Statements
```

Calling the functions.

```
fun_name()
```

ex -

```
def add1():
    a=int(input("enter a value"))
    b=int(input("enter b value"))
    s=a+b
    print(f"the sum of {a} and {b} is {s}")
add1()
```

o/p->

enter a value 34

enter b value 12

the sum of 34 and 12 is 46

the variable inside the function is called local variable.

↳ the variable inside the function

here we cant get {a} value because it is inside the function.

local variable: Any variable declared inside function called as local variable.

Global variable: Any variable declared outside the function is called as global variable.

2) function with input and without return -

syntax def fun_name (p1, p2..., pn):
    Statements

calling the function -

read the parameters p1, p2.., pn # first we have to read the Parameters.

fun_name(p1, p2..., pn) # next we have to call the function name -

ex - def add2(n1, n2):
        s = n1+n2
        print f" the sum of {n1} and {n2} is {s}")

add2(5, 8)

the sum of 5 and 8 is 13.

3) function without input and with return -

syntax - def fun_name():
            Statements
            return value

calling the function

var = fun_name()

Statements

ex - def add3():
        a = int(input("enter a value"))
        b = int(input("enter b value"))
        s = a+b
        return a, b, s

x, y, z = add3()

enter a value 12

enter b value 8

print(f"the sum of {x} and {y} is {z}")

the sum of 12 and 8 is 20

4) function with input & with return -

syntax-

```
def fun_name(P1, P2 ...., Pn ):
    statements
    return value
```

Calling the function -

```
read the parameters P1, P2 ..., Pn
Var =fun_name (P1, P2 ..., Pn)
statements.
```

ex -

```
def add4(num1, num2):
    s=num1+num2
    return s

x=12
y=15
sum_out =add4(x, y)
print(f"the sum of {x} and {y} is {sum_out}")
```

the sum of 12 and 15 is 27.

10) Check given number is prime or not with method1.

```
def prime1( ):
    num =int(input("enter a value"))
    if num<=1:
            print(num,"is not a prime number")
    else:
        for i in range (2,num,1):
                if (num%i==0):
                        print(num," is not a prime number")
                        break
        else:
            print(num, "is a prime number")
```

2) Calculate factorial of given num with method 2.

```
def fact1(n):
    f=1
    for i in range(1,n+1,1):
        f=f*i
    print(f"factorial of {n} is {f}")

fact1(5)
```

factorial of 5 is 120.

3) Sum of number, sum of even number, sum of odd numb of given range with method 4.

```
def sum_num(P1,P2):
    ts=0
    es=0
    os=0
    for i in range(P1,P2+1,1):
        ts =ts+i
        if (i%2==0):
            es =es+i
        else:
            os =os+i
    return ts,es,os

s,e,od = sum_num(1,10)

print(f"the total sum is {s}\neven sum is {e}\nodd sum is {od}
```

the total sum is 55

even sum is 30

odd sum is 25